

# Commutative $m$ -ary Quasigroup Actions in a 2-adic Regime: A CSIDH-like Commutative-Action Candidate and a Key Exchange Protocol

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**Abstract:** Isogeny-based cryptography has popularized commutative-action key exchange (e.g., CSIDH-style class-group actions) as a post-quantum analogue of Diffie–Hellman.[DF17, CLM<sup>+</sup>18] We propose *Commutative Quasigroup Actions (CQA)*: a fully general  $m$ -ary ( $m \geq 3$ ) quasigroup-inspired commutative action over  $\mathbb{Z}/2^w\mathbb{Z}$  whose commutativity holds in a *2-adic regime* motivated by polynomial permutations on the units of  $\mathbb{Z}_{2^w}$ . [MŠG10] The construction uses  $m$  structured odd-degree polynomials with explicit parity constraints (even base point  $g$ , odd secret components) and a cyclic wiring that reuses the secret vector under rotation. Extensive numerical validation (2D parameter sweeps over base width  $w$ , 2-adic valuation  $v_2(g)$ , rounds  $R$ , and polynomial degree) shows that:

- the effective image size of both the public key and the shared secret shrinks as a power of two with  $v_2(g)$ ;
- for all safe parameter regions ( $w \geq 40$  or  $v_2(g) \geq 3$ ) the sets of public keys and shared secrets are *completely disjoint*;
- odd round counts often yield approximately twice the image size of even counts (“lucky rounds”).

These properties provide strong evidence that CQA realizes a large functional graph on its effective image while keeping the public-key orbit and the shared-secret orbit in algebraically separated regions — exactly the behaviour required for a secure commutative-action KEX. We describe the construction, connect it to commutative-action thinking in isogeny cryptography, prove a 1-dimensional collapse attack when  $g$  is odd, and list concrete open problems.

**1. Construction: general  $m$ -ary cyclic quasigroup action.** Fix an arity  $m \geq 3$  and a modulus  $2^w$  (with  $w$  a security parameter). A *family*  $\mathcal{F} = (g, (P_1, \dots, P_m))$  consists of a public base point  $g \in \mathbb{Z}/2^w\mathbb{Z}$  and  $m$  structured odd-degree polynomials  $P_1, \dots, P_m$  over  $\mathbb{Z}/2^w\mathbb{Z}$ . Each polynomial is required to satisfy explicit parity constraints that enforce the 2-adic regime:

- the constant term of  $g$  is even ( $g = 2s$ );
- all secret components (the vector  $K = (k_1, \dots, k_{m-1})$ ) are odd ( $k_i = 2s_i + 1$ );
- the polynomials  $P_j$  have odd degree and coefficients compatible with the above parity (leading coefficient even for the  $g$ -slot, odd otherwise).

These constraints are not arbitrary; they are the minimal conditions under which the cyclic action commutes in the 2-adic topology (see Section 1.1).

*The local law.* The elementary operation is the multilinear-like product

$$f(x_1, \dots, x_m) := \prod_{i=1}^m P_i(x_i) \pmod{2^w}.$$

Because each  $P_i$  has odd degree,  $f$  is a permutation polynomial on the units of  $\mathbb{Z}_{2^w}$  when the arguments are units; the even/odd parity constraints extend this behaviour to the full ring in a controlled way.

*Cyclic wiring (macro-step).* Let  $K = (k_1, \dots, k_{m-1})$  be the secret vector and let  $x$  be the current state. A single *macro-step*  $M_K(x)$  is defined by the following cyclic insertion procedure (exactly as implemented in the reference Python proof-of-concept):

1. Start with the length- $m$  argument vector  $\mathbf{a}^{(0)} = (k_1, \dots, k_{m-1}, x)$ .
2. For  $i = 0, \dots, m - 2$ :
  - Compute  $y_{i+1} = f(\mathbf{a}^{(i)})$ .
  - Rotate the secret part of  $\mathbf{a}^{(i)}$  one position to the left and insert  $y_{i+1}$  into the vacated slot, obtaining  $\mathbf{a}^{(i+1)}$ .
3. Return  $y_{m-1}$  as the output of the macro-step.

The full action is the  $R$ -fold iteration:

$$\text{Act}(K, x) := M_K^R(x).$$

For  $m = 3$  the procedure specialises exactly to the three-micro-step wiring given in the earlier draft:

$$y_1 = f(a, b, x), \quad y_2 = f(b, y_1, a), \quad y_3 = f(y_2, a, b), \quad M_{(a,b)}(x) = y_3.$$

**1.1. Design rationale: necessity of an even base point  $g$ .** The parity constraints are not merely implementation details; they are required to avoid a fatal one-dimensional collapse. Consider the homogeneous specialisation in which all constant terms of the  $P_j$  vanish and  $g$  is odd (i.e., a unit in  $\mathbb{Z}_{2^w}$ ). Then the entire macro-step collapses to pure scalar multiplication:

$$M_K(x) = \mu(K) \cdot x, \quad \mu(K) \in \mathbb{Z}/2^w\mathbb{Z}.$$

(The scalar  $\mu(K)$  is a monomial in the components of  $K$  whose precise form depends on  $m$  and the degrees of the  $P_j$ .) Consequently the public key becomes

$$A \cdot g = \mu(A) \cdot g.$$

Because  $g$  is invertible, an adversary who sees  $A \cdot g$  can immediately recover the full action scalar:

$$\mu(A) = (A \cdot g) \cdot g^{-1}.$$

He can then compute the shared secret without ever learning  $A$  or  $B$ :

$$\text{ss} = \mu(A) \cdot (B \cdot g) = \mu(A) \cdot \mu(B) \cdot g.$$

This is precisely the “hidden homogeneous space” attack that renders any one-dimensional commutative action insecure. The requirement that  $g$  be even (together with odd secret components and odd-degree polynomials) prevents the homogeneous collapse and forces the action to explore a genuinely higher-dimensional subset of the ring. Extensive numerical experiments confirm that, once  $v_2(g)$  is large enough relative to  $w$  and  $R$ , the resulting functional graph on the effective image is large and the public-key and shared-secret orbits remain disjoint — the algebraic signature of a secure commutative action.

**2. KEX protocol (CSIDH-like commutative action).** Public parameters are a family  $\mathcal{F} = (g, (P_1, \dots, P_m))$  together with the round count  $R$ . Each party samples a secret vector ( $A$  for Alice,  $B$  for Bob) whose components satisfy the odd-parity constraint and publishes

$$A \cdot g := \text{Act}(A, g), \quad B \cdot g := \text{Act}(B, g).$$

The shared secret is computed as

$$\text{ss} := \text{Act}(A, B \cdot g) = \text{Act}(B, A \cdot g),$$

where equality holds throughout the safe 2-adic regime (empirically verified for thousands of random instances across  $w \in \{32, 36, 40, 44, 48\}$ ,  $v_2(g) \in \{1, \dots, 18\}$ ,  $R \in \{2, \dots, 16\}$ , and polynomial degrees 3 and 9). A standard KDF (e.g., HKDF-SHA-256) is applied to  $\text{ss}$  to derive session keys.

**3. Communication sizes and effective entropy.** A public key is a single element of  $\mathbb{Z}/2^w\mathbb{Z}$  and therefore costs  $w$  bits ( $\lceil w/8 \rceil$  bytes). The KEX transcript consists of two public keys plus optional KDF context. A secret key is a vector of  $m - 1$  odd elements mod  $2^w$ ; a naïve encoding costs  $(m - 1)w$  bits. In the 2-adic regime the distribution of each secret component has a frozen low-order suffix whose length grows with  $v_2(g)$  and  $R$ . Consequently the *effective* entropy per component is approximately

$$\text{eff.bits} \approx w - v_2(g) - c \cdot R$$

for a small constant  $c$  that depends on the polynomial degree (empirically  $c \approx 1$  for degree 3 and slightly larger for degree 9). The 2D sweep data show that safe parameter choices (those that keep  $\text{Apub}$  and  $\text{ssA}$  disjoint) still leave  $\text{eff.bits} \geq w/2$  for  $w \geq 40$  and modest  $R$ , which is sufficient for conservative  $\lambda$ -bit security against generic collision search.

**4. Empirical validation (2D parameter sweeps).** We performed exhaustive adaptive 2D sweeps over the rectangle

$$w \in \{32, 36, 40, 44, 48\}, \quad v_2(g) \in \{1, \dots, 18\}, \quad R \in \{2, \dots, 16\}$$

for both degree-3 and degree-9 odd polynomials (more than 2700 distinct cells, each sampled until saturation or a  $10^7$  sample cap). The results reveal three robust statistical laws:

1. **Power-of-two image sizes.** Both  $|\text{image}(\text{Apub})|$  and  $|\text{image}(\text{ssA})|$  are always exact powers of two. The size halves (to within  $\pm 1$ ) with every increment of  $v_2(g)$ .
2. **“Lucky rounds”.** For any fixed  $(w, v_2(g))$ , an odd number of rounds yields an image approximately twice as large as the nearest even count. This periodic mixing gain is reproducible across all tested widths and valuations.
3. **Strong set separation.** For every cell with  $w \geq 40$  or  $v_2(g) \geq 3$  we observed

$$|\text{Apub} \cap \text{ssA}| = 0$$

(perfect algebraic separation). The only exceptions form a narrow “bad zone” confined to  $(w = 32, v_2(g) \leq 2)$  and, for higher-degree polynomials, a slightly larger but still tiny region around  $(w = 36, v_2(g) = 2)$ . In all bad-zone cells the overlap is exactly 100% ( $\text{Apub} = \text{ssA}$ ); outside it the overlap is exactly 0%.

These laws confirm that the construction realises a large functional graph on its effective image while automatically placing the public-key orbit and the shared-secret orbit in disjoint subsets — the precise algebraic property needed for a secure commutative-action KEX.

## 5. Open research questions.

1. **Hardness assumptions.** The natural computational problems are:
  - **Vectorization / key recovery.** Given  $\mathcal{F}$ ,  $g$  and  $y = \text{Act}(K, g)$ , recover a vector  $K'$  with  $\text{Act}(K', g) = y$ .
  - **Shared-secret computation.** Given  $g$ ,  $A \cdot g$  and  $B \cdot g$ , compute  $\text{ss} = \text{Act}(A, B \cdot g)$  without recovering  $A$  or  $B$ .
  - **Distinguishing.** Distinguish the distribution of  $\text{ss}$  from uniform on the observed image (or uniform modulo the frozen suffix).

The cyclic wiring together with the nonlinear odd-degree polynomials and the even- $g$  constraint make reduction to a single “missing quasigroup argument” impossible; the 1D collapse attack is provably blocked. Whether these problems are hard in the 2-adic regime remains the central open question.

2. **Characterisation of the commutative 2-adic subfamily.** Why exactly does the parity-constrained family commute? Can one give an algebraic characterisation of the maximal commutative subfamily, or an efficient algorithm that decides membership?
3. **Parameter selection (conservative profiles).** Combining the image-size law with the separation guarantee, we propose the following draft tiers (all with  $m = 3$ , degree 3,  $R = 2$  or 3):
  - **128-bit tier:**  $w = 512$  ( $v_2(g) = 4..6$ ), public key 64 bytes, KEX transcript 128 bytes.
  - **192-bit tier:**  $w = 768$  ( $v_2(g) = 5..7$ ), public key 96 bytes, KEX 192 bytes.
  - **256-bit tier:**  $w = 1024$  ( $v_2(g) = 6..8$ ), public key 128 bytes, KEX 256 bytes.

These choices keep the construction comfortably inside the perfect-separation region while retaining  $\text{eff.bits} \geq 2\lambda$ .

4. **Security models (CPA/CCA and AKE).** The current Python proof-of-concept validates commutativity, serialization and KDF usage in the passive setting. Upgrading to active security (IND-CCA KEM, AKE) is an open engineering and proof challenge.
5. **Attacks and leakage.** Identify what structure is necessarily leaked by the 2-adic regime (frozen suffix, image-size law) and how KDF usage, masking, or protocol-level choices mitigate it.
6. **Signatures and identification.** Explore whether CQA admits identification/signature analogues of group-action signatures without breaking commutativity.

## 1 Symbolic affine commutator calculations

The following symbolic derivation shows how a 3-ary affine model becomes a unary affine self-map and why affine commutativity is too weak to be cryptographically interesting. (The same expansion works verbatim for general  $m$  with only notational changes.)

We work in the  $m$ -ary cyclic wiring with

$$f(u_1, \dots, u_m) = P_1(u_1) \cdots P_m(u_m)$$

and affine slot polynomials

$$P_j(t) = c_j + d_j t, \quad j = 1, \dots, m.$$

**Proposition 1.1** (Affine  $m$ -ary macro-step). *Under the affine truncation the full macro-step is affine in the state variable:*

$$M_K(x) = \alpha(K) + \beta(K)x.$$

The explicit formulae for  $\alpha$  and  $\beta$  are obtained by tracking the constant term and the coefficient of  $x$  through the  $m - 1$  micro-steps; they are multilinear in the secret components and involve only the constants  $c_j$  and the slopes  $d_j$ .

**Proposition 1.2** (Commutator obstruction in the affine regime). *Let  $A$  and  $B$  be two secret vectors and write  $M_A(x) = \alpha_A + \beta_A x$ ,  $M_B(x) = \alpha_B + \beta_B x$ . Then*

$$M_A(M_B(x)) - M_B(M_A(x)) = \alpha_A(1 - \beta_B) - \alpha_B(1 - \beta_A).$$

*In particular, affine-slot multilinearity alone does not imply useful commutativity.*

**Proposition 1.3** (Homogeneous specialization). *If all constants vanish ( $c_j \equiv 0$ ) then the macro-step collapses to pure scalar multiplication*

$$M_K(x) = \left( \prod_{j=1}^m d_j \right) \left( \prod_{i=1}^{m-1} k_i^d \right) x$$

*for a monomial exponent  $d$  that depends only on the polynomial degrees. Two such scalar maps commute automatically, but the construction is exactly the one-dimensional case shown to be insecure when  $g$  is odd (Section 1.1).*

## 2 Conclusion

The generic  $m$ -ary construction, the explicit parity constraints, the even- $g$  requirement, and the extensive numerical validation together give a concrete, implementable candidate for a post-quantum commutative-action key exchange whose security rests on the hardness of vectorization and shared-secret computation in a genuinely high-dimensional 2-adic regime. The only remaining task is to turn the empirical evidence into a rigorous hardness proof — or to discover an attack that exploits the observed image-size law.

## References

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